

Assignment 7: GLMs (Linear Regressios, ANOVA, & t-tests)

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OVERVIEW

This exercise accompanies the lessons in Environmental Data Analytics on generalized linear models.

Directions

1. Rename this file `<FirstLast>_A07_GLMs.Rmd` (replacing `<FirstLast>` with your first and last name).
2. Change “Student Name” on line 3 (above) with your name.
3. Work through the steps, **creating code and output** that fulfill each instruction.
4. Be sure to **answer the questions** in this assignment document.
5. When you have completed the assignment, **Knit** the text and code into a single PDF file.

!! AI generated code should NOT be used in this assignment

Set up your session

1. Set up your session. Check your working directory. Load the tidyverse, agricolae and other needed packages. Import the *raw* NTL-LTER raw data file for chemistry/physics (NTL-LTER_Lake_ChemistryPhysics_Raw.csv). Set date columns to date objects.
2. Build a ggplot theme and set it as your default theme.

#1

```
library(tidyverse)
```

```
## -- Attaching core tidyverse packages ----- tidyverse 2.0.0 --
## v dplyr      1.1.4      v readr      2.1.5
## v forcats    1.0.0      v stringr   1.5.1
## v ggplot2    4.0.0      v tibble    3.2.1
## v lubridate  1.9.4      v tidyr     1.3.1
## v purrr      1.0.2
```

```
## -- Conflicts ----- tidyverse_conflicts() --
```

```
## x dplyr::filter() masks stats::filter()
```

```
## x dplyr::lag()     masks stats::lag()
```

```
## i Use the conflicted package (<http://conflicted.r-lib.org/>) to force all conflicts to become errors
```

```
install.packages("agricolae")
```

```
## Installing package into '/home/guest/R/x86_64-pc-linux-gnu-library/4.4'
## (as 'lib' is unspecified)
```

```
library(agricolae)
library(lubridate)
getwd()
```

```
## [1] "/home/guest/EDE_Fall2025"
```

```
setwd("/home/guest/EDE_Fall2025/Data/Raw")
data1 <- read_csv("NTL-LTER_Lake_ChemistryPhysics_Raw.csv")
```

```
## Rows: 38614 Columns: 11
## -- Column specification -----
## Delimiter: ","
## chr (4): lakeid, lakename, sampleddate, comments
## dbl (7): year4, daynum, depth, temperature_C, dissolvedOxygen, irradianceWat...
##
## i Use 'spec()' to retrieve the full column specification for this data.
## i Specify the column types or set 'show_col_types = FALSE' to quiet this message.
```

```
#2
theme_set(theme_bw())
```

Simple regression

Our first research question is: Does mean lake temperature recorded during July change with depth across all lakes?

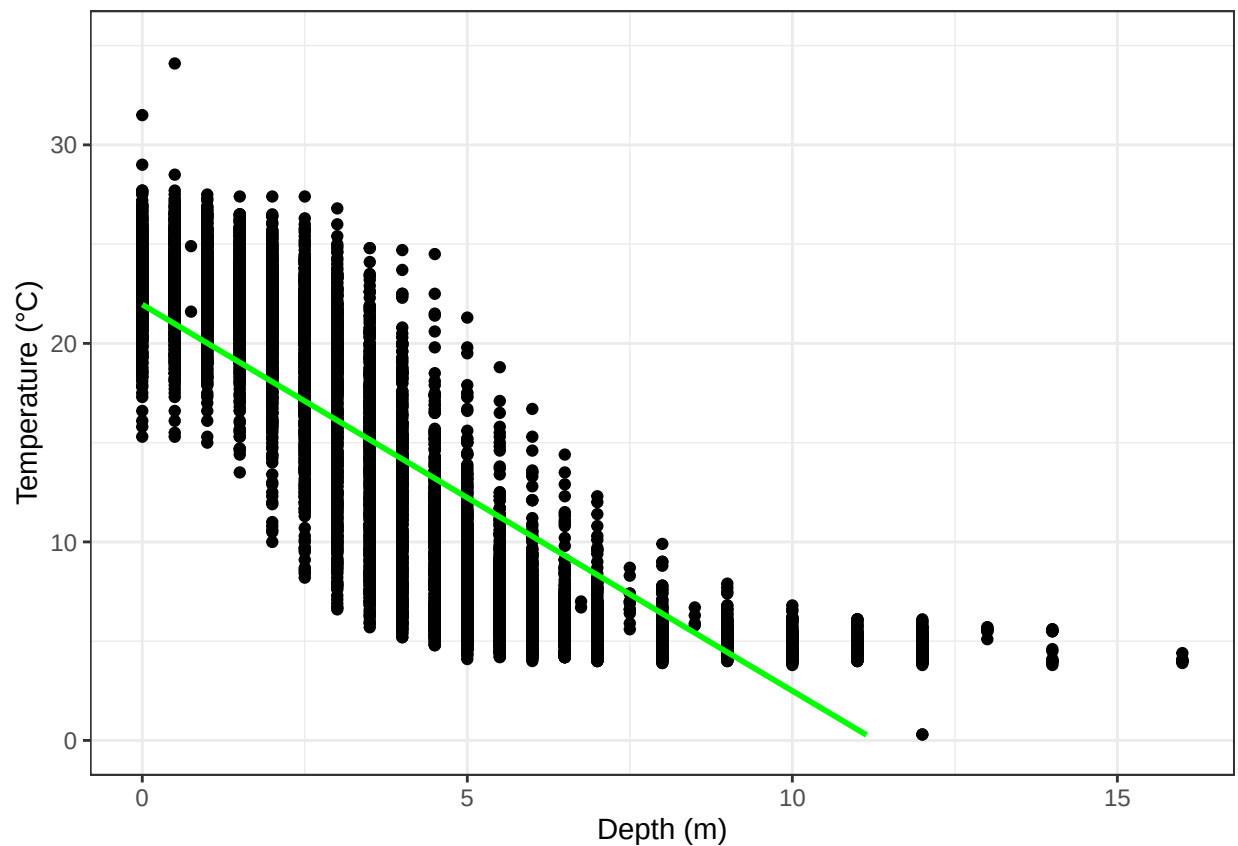
3. State the null and alternative hypotheses for this question: > Answer: H0: Mean temperature does not change with depth. Ha: Mean temperature does change with depth.
4. Wrangle your NTL-LTER dataset with a pipe function so that the records meet the following criteria:
 - Only dates in July.
 - Only the columns: lakename, year4, daynum, depth, temperature_C
 - Only complete cases (i.e., remove NAs)
5. Visualize the relationship among the two continuous variables with a scatter plot of temperature by depth. Add a smoothed line showing the linear model, and limit temperature values from 0 to 35 °C. Make this plot look pretty and easy to read.

```
#4
data1$sampledate <- as.Date(data1$sampledate, format = "%m/%d/%y")
plot_july <- data1 %>%
  filter(month(sampledate)==7) %>%
  select("lakename", "year4", "daynum", "depth", "temperature_C") %>%
  drop_na()
#5
ggplot(plot_july, aes(x = depth, y = temperature_C)) +
```

```
geom_point() +
geom_smooth(method = "lm", se = FALSE, color = "green") +
ylim(0, 35) +
labs(x = "Depth (m)", y = "Temperature (°C)") +
theme_bw()
```

```
## 'geom_smooth()' using formula = 'y ~ x'
```

```
## Warning: Removed 24 rows containing missing values or values outside the scale range
## ('geom_smooth()').
```



6. Interpret the figure. What does it suggest with regards to the response of temperature to depth? Do the distribution of points suggest about anything about the linearity of this trend?

Answer: Lake temperature decreases as depth increases. This would be considered a negative relationship. The distribution of points descends downward in a somewhat straight pattern which would suggest a rough linear relationship.

7. Perform a linear regression to test the relationship and display the results.

```
#7
lr_temp_depth <- lm(temperature_C ~ depth, data = plot_july)
summary(lr_temp_depth)
```

```
##
## Call:
## lm(formula = temperature_C ~ depth, data = plot_july)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -9.5173 -3.0192  0.0633  2.9365 13.5834
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept) 21.95597    0.06792   323.3  <2e-16 ***
## depth       -1.94621    0.01174  -165.8  <2e-16 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 3.835 on 9726 degrees of freedom
## Multiple R-squared:  0.7387, Adjusted R-squared:  0.7387
## F-statistic: 2.75e+04 on 1 and 9726 DF,  p-value: < 2.2e-16
```

8. Interpret your model results in words. Include how much of the variability in temperature is explained by changes in depth, the degrees of freedom on which this finding is based, and the statistical significance of the result. Also mention how much temperature is predicted to change for every 1m change in depth.

Answer: This linear regression demonstrates a strong statistical relationship between lake temperature and depth for the month of July. This model explains about 73.9% of the variability in temperature across all samples (“R squared =0.7397 which is based on 1 and 9726 degrees of freedom”). The given p-value (“ $<2 \times 10^{-16}$ ”) means the relationship is highly significant with depth being a strong prediction for temperature. The slope is approximately -1.946. This means that for every 1m increase in depth, water temperature decreases by 1.95 degrees C. Therefore, the deeper the water, the colder the temperature.

Multiple regression

Let’s tackle a similar question from a different approach. Here, we want to explore what might the best set of predictors for lake temperature in July across the monitoring period at the North Temperate Lakes LTER.

9. Run an AIC to determine what set of explanatory variables (year4, daynum, depth) is best suited to predict temperature.
10. Run a multiple regression on the recommended set of variables.

```
#9
t1 <- lm(temperature_C ~ depth, data = plot_july)
t2 <- lm(temperature_C ~ depth + year4, data = plot_july)
t3 <- lm(temperature_C ~ depth + daynum, data = plot_july)
t4 <- lm(temperature_C ~ depth + year4 + daynum, data = plot_july)
AIC(t1, t2, t3, t4)
```

```
##      df      AIC
## t1  3 53762.12
## t2  4 53756.97
## t3  4 53679.36
## t4  5 53674.39
```

```
#10
recommended_model <- lm(temperature_C ~ depth + year4 + daynum, data = plot_july)
summary(recommended_model)
```

```
##
## Call:
## lm(formula = temperature_C ~ depth + year4 + daynum, data = plot_july)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -9.6536 -3.0000  0.0902  2.9658 13.6123
##
## Coefficients:
##              Estimate Std. Error  t value Pr(>|t|)
## (Intercept) -8.575564   8.630715  -0.994  0.32044
## depth       -1.946437   0.011683 -166.611 < 2e-16 ***
## year4        0.011345   0.004299   2.639  0.00833 **
## daynum       0.039780   0.004317   9.215 < 2e-16 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 3.817 on 9724 degrees of freedom
## Multiple R-squared:  0.7412, Adjusted R-squared:  0.7411
## F-statistic: 9283 on 3 and 9724 DF,  p-value: < 2.2e-16
```

11. What is the final set of explanatory variables that the AIC method suggests we use to predict temperature in our multiple regression? How much of the observed variance does this model explain? Is this an improvement over the model using only depth as the explanatory variable?

Answer: The final set of variables included depth, year, and day number for the predictors for July lake temperature. There is a slight increase in variation for temperature than the original regression using depth, the R squared is 0.7412 which is barely above 0.74. It is an improvement. Depth is still the dominant variable but there are smaller yet determinable effects with sampling within the year.

Analysis of Variance

12. Now we want to see whether the different lakes have, on average, different temperatures in the month of July. Run an ANOVA test to complete this analysis. (No need to test assumptions of normality or similar variances.) Create two sets of models: one expressed as an ANOVA models and another expressed as a linear model (as done in our lessons).

#12

```
anova_model <- aov(temperature_C ~ lakenname, data = plot_july)
summary(anova_model)
```

```
##              Df Sum Sq Mean Sq F value Pr(>F)
## lakenname      8  21642   2705.2     50 <2e-16 ***
## Residuals    9719 525813     54.1
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

```
lm_model <- lm(temperature_C ~ lakenname, data = plot_july)
summary(lm_model)
```

```
##
## Call:
## lm(formula = temperature_C ~ lakenname, data = plot_july)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -10.769  -6.614  -2.679   7.684  23.832
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept)    17.6664     0.6501  27.174 < 2e-16 ***
## lakennameCrampton Lake    -2.3145     0.7699   -3.006 0.002653 **
## lakennameEast Long Lake   -7.3987     0.6918 -10.695 < 2e-16 ***
## lakennameHummingbird Lake -6.8931     0.9429  -7.311 2.87e-13 ***
## lakennamePaul Lake       -3.8522     0.6656  -5.788 7.36e-09 ***
## lakennamePeter Lake      -4.3501     0.6645  -6.547 6.17e-11 ***
## lakennameTuesday Lake    -6.5972     0.6769  -9.746 < 2e-16 ***
## lakennameWard Lake       -3.2078     0.9429  -3.402 0.000672 ***
## lakennameWest Long Lake  -6.0878     0.6895  -8.829 < 2e-16 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 7.355 on 9719 degrees of freedom
## Multiple R-squared:  0.03953,    Adjusted R-squared:  0.03874
## F-statistic:    50 on 8 and 9719 DF,  p-value: < 2.2e-16
```

13. Is there a significant difference in mean temperature among the lakes? Report your findings.

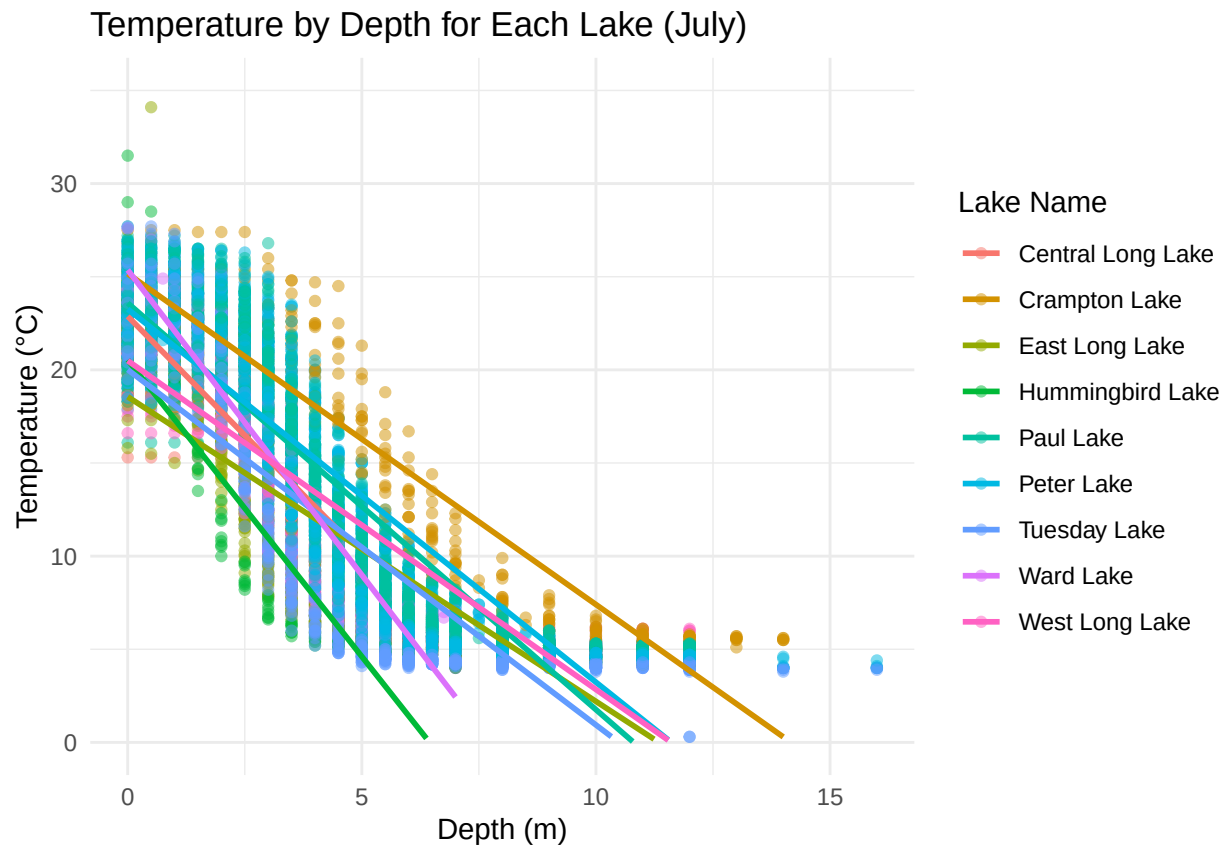
Answer: There is a serious statistical difference in mean July temperature among lakes. ANOVA results are: F-value is 8, 9719 = 50, $p < 2 \times 10^{-16}$ which indicates at least one lake mean temp is different than the rest. This is supported by the linear model output. Several lakes have significantly different mean temperatures as compared to Crystal Lake. I believe Crystal Lake is meant to be the reference lake. Some data of note, East Long Lake/Tuesday Lake and West Long Lake have significantly different mean temperatures compared to Crystal Lake, as $p < 0.001$. Total variation is 0.0395 but still statistically significant.

14. Create a graph that depicts temperature by depth, with a separate color for each lake. Add a `geom_smooth` (method = "lm", se = FALSE) for each lake. Make your points 50 % transparent. Adjust your y axis limits to go from 0 to 35 degrees. Clean up your graph to make it pretty.

```
#14.
ggplot(plot_july, aes(x = depth, y = temperature_C, color = lakename)) +
  geom_point(alpha = 0.5) +
  geom_smooth(method = "lm", se = FALSE) +
  ylim(0, 35) +
  labs(
    x = "Depth (m)",
    y = "Temperature (°C)",
    color = "Lake Name",
    title = "Temperature by Depth for Each Lake (July)"
  ) +
  theme_minimal()
```

```
## 'geom_smooth()' using formula = 'y ~ x'
```

```
## Warning: Removed 73 rows containing missing values or values outside the scale range
## ('geom_smooth()').
```



15. Use the Tukey's HSD test to determine which lakes have different means.

```
#15
tukey_test <- TukeyHSD(anova_model)
tukey_test
```

```
## Tukey multiple comparisons of means
## 95% family-wise confidence level
##
## Fit: aov(formula = temperature_C ~ lakename, data = plot_july)
##
## $lakename
##
```

	diff	lwr	upr	p adj
## Crampton Lake-Central Long Lake	-2.3145195	-4.7031913	0.0741524	0.0661566
## East Long Lake-Central Long Lake	-7.3987410	-9.5449411	-5.2525408	0.0000000
## Hummingbird Lake-Central Long Lake	-6.8931304	-9.8184178	-3.9678430	0.0000000
## Paul Lake-Central Long Lake	-3.8521506	-5.9170942	-1.7872070	0.0000003
## Peter Lake-Central Long Lake	-4.3501458	-6.4115874	-2.2887042	0.0000000
## Tuesday Lake-Central Long Lake	-6.5971805	-8.6971605	-4.4972005	0.0000000
## Ward Lake-Central Long Lake	-3.2077856	-6.1330730	-0.2824982	0.0193405
## West Long Lake-Central Long Lake	-6.0877513	-8.2268550	-3.9486475	0.0000000
## East Long Lake-Crampton Lake	-5.0842215	-6.5591700	-3.6092730	0.0000000
## Hummingbird Lake-Crampton Lake	-4.5786109	-7.0538088	-2.1034131	0.0000004
## Paul Lake-Crampton Lake	-1.5376312	-2.8916215	-0.1836408	0.0127491
## Peter Lake-Crampton Lake	-2.0356263	-3.3842699	-0.6869828	0.0000999
## Tuesday Lake-Crampton Lake	-4.2826611	-5.6895065	-2.8758157	0.0000000
## Ward Lake-Crampton Lake	-0.8932661	-3.3684639	1.5819317	0.9714459
## West Long Lake-Crampton Lake	-3.7732318	-5.2378351	-2.3086285	0.0000000
## Hummingbird Lake-East Long Lake	0.5056106	-1.7364925	2.7477137	0.9988050
## Paul Lake-East Long Lake	3.5465903	2.6900206	4.4031601	0.0000000
## Peter Lake-East Long Lake	3.0485952	2.2005025	3.8966879	0.0000000
## Tuesday Lake-East Long Lake	0.8015604	-0.1363286	1.7394495	0.1657485
## Ward Lake-East Long Lake	4.1909554	1.9488523	6.4330585	0.0000002
## West Long Lake-East Long Lake	1.3109897	0.2885003	2.3334791	0.0022805
## Paul Lake-Hummingbird Lake	3.0409798	0.8765299	5.2054296	0.0004495
## Peter Lake-Hummingbird Lake	2.5429846	0.3818755	4.7040937	0.0080666
## Tuesday Lake-Hummingbird Lake	0.2959499	-1.9019508	2.4938505	0.9999752
## Ward Lake-Hummingbird Lake	3.6853448	0.6889874	6.6817022	0.0043297
## West Long Lake-Hummingbird Lake	0.8053791	-1.4299320	3.0406903	0.9717297
## Peter Lake-Paul Lake	-0.4979952	-1.1120620	0.1160717	0.2241586
## Tuesday Lake-Paul Lake	-2.7450299	-3.4781416	-2.0119182	0.0000000
## Ward Lake-Paul Lake	0.6443651	-1.5200848	2.8088149	0.9916978
## West Long Lake-Paul Lake	-2.2356007	-3.0742314	-1.3969699	0.0000000
## Tuesday Lake-Peter Lake	-2.2470347	-2.9702236	-1.5238458	0.0000000
## Ward Lake-Peter Lake	1.1423602	-1.0187489	3.3034693	0.7827037
## West Long Lake-Peter Lake	-1.7376055	-2.5675759	-0.9076350	0.0000000
## Ward Lake-Tuesday Lake	3.3893950	1.1914943	5.5872956	0.0000609
## West Long Lake-Tuesday Lake	0.5094292	-0.4121051	1.4309636	0.7374387
## West Long Lake-Ward Lake	-2.8799657	-5.1152769	-0.6446546	0.0021080

16. From the findings above, which lakes have the same mean temperature, statistically speaking, as Peter Lake? Does any lake have a mean temperature that is statistically distinct from all the other lakes?

Answer: Peter Lake has statistically similar mean July temperatures to Paul Lake and Ward Lake. The rest of the lakes are significantly different from Peter Lake with p being < 0.05 . There were no lakes that was significantly different from all the other lakes. It is worth noting that East Long Lake is extremely cold and does differ from many, but not all, of the lakes.

17. If we were just looking at Peter Lake and Paul Lake. What's another test we might explore to see whether they have distinct mean temperatures?

Answer: You could use an independent samples t-test to directly compare the mean temperature of both groups to see if there is a statistically significant difference between them.

18. Wrangle the July data to include only records for Crampton Lake and Ward Lake. Run the two-sample T-test on these data to determine whether their July temperature are same or different. What does the test say? Are the mean temperatures for the lakes equal? Does that match you answer for part 16?

```
cramptonandward_lake <- subset(plot_july, lakename %in% c("Crampton Lake",  
  "Ward Lake"))  
t.test(temperature_C ~ lakename, data = cramptonandward_lake)
```

```
##  
## Welch Two Sample t-test  
##  
## data: temperature_C by lakename  
## t = 1.1181, df = 200.37, p-value = 0.2649  
## alternative hypothesis: true difference in means between group Crampton Lake and group Ward Lake is not equal to 0  
## 95 percent confidence interval:  
## -0.6821129 2.4686451  
## sample estimates:  
## mean in group Crampton Lake mean in group Ward Lake  
## 15.35189 14.45862
```

Answer: The two-sample t-test comparing both lakes shows a p-value greater than 0.05 therefore, the difference in mean July temperatures is not statistically significant. I fail to reject the null hypothesis as the mean temperatures for both lakes is statistically equal. This result matches the previous answer in Q16 as the Tukey test indicated no significant difference between both lakes.